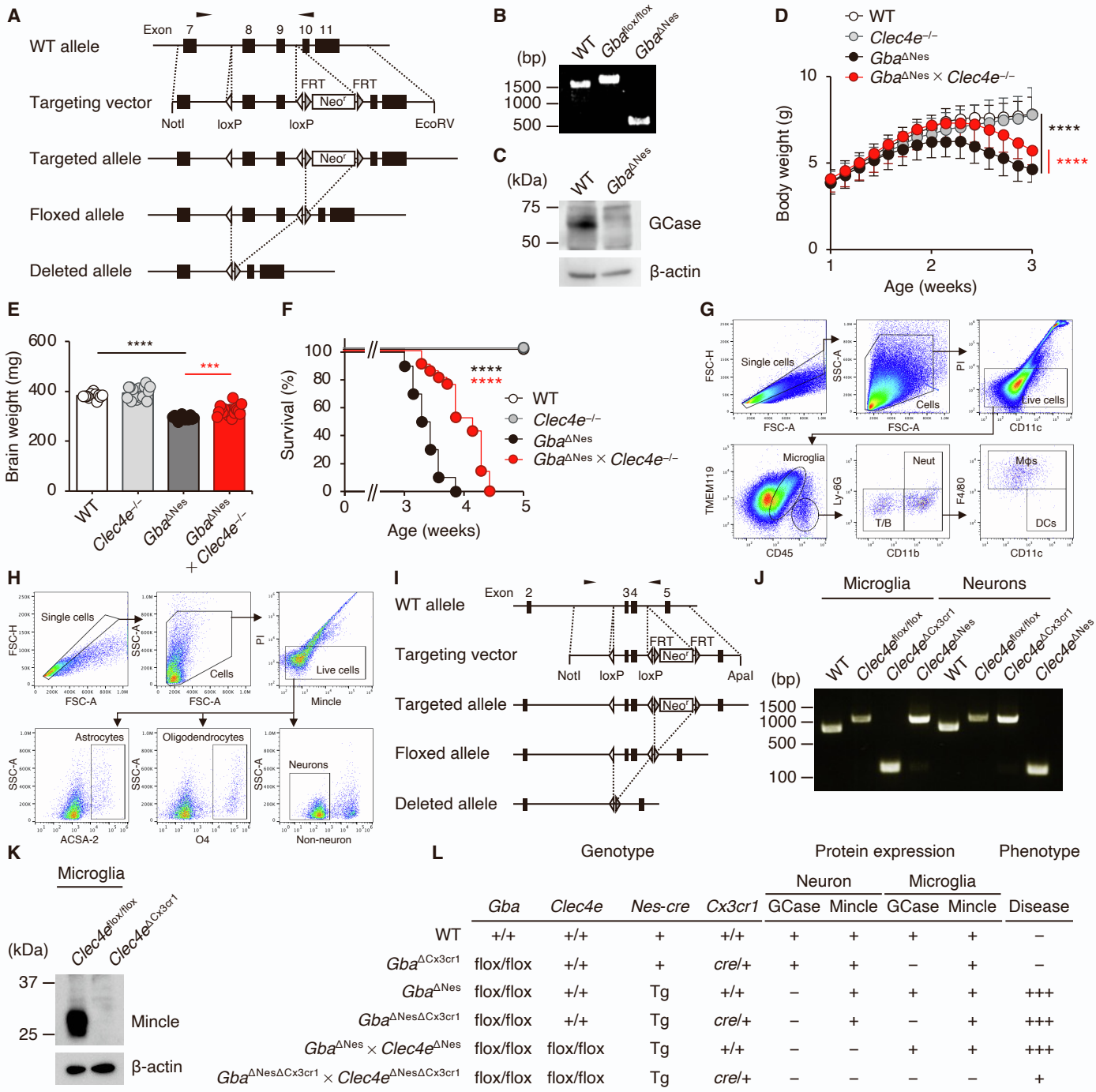
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Supplemental information

Direct activation of microglia by β-glucosylceramide causes phagocytosis of neurons that exacerbates Gaucher disease

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	Genotype				Protein expression				Phenotype
					Neuron		Microglia		
	Gba	*Clec4e*	*Nes-cre*	*Cx3cr1*	GCase	Mincle	GCase	Mincle	Disease
WT	+/+	+/+	+	+/+	+	+	+	+	–
Gba$^{\Delta Cx3cr1}$	flox/flox	+/+	+	*cre*/+	+	+	–	+	–
Gba$^{\Delta Nes}$	flox/flox	+/+	Tg	+/+	–	+	+	+	+++
Gba$^{\Delta Nes\Delta Cx3cr1}$	flox/flox	+/+	Tg	*cre*/+	–	+	–	+	+++
Gba$^{\Delta Nes}$ × *Clec4e*$^{\Delta Nes}$	flox/flox	flox/flox	Tg	+/+	–	–	+	+	+++
Gba$^{\Delta Nes\Delta Cx3cr1}$ × *Clec4e*$^{\Delta Nes\Delta Cx3cr1}$	flox/flox	flox/flox	Tg	*cre*/+	–	–	–	–	+

Figure S1. Characterization of mouse GD models (Related to Figure 1)

(A) Schematic view of the generation of *Gba*$^{\Delta Nes}$ mice. Black bars represent *Gba* exons, open triangles represent loxP sites, gray triangles represent flippase recognition target (FRT) sites, and arrow heads denote primer location. (B) PCR products showing removal of the exons 8 and 9 of *Gba* in the brain of *Gba*$^{\Delta Nes}$ mice. (C) Western blot analysis of GCase expression in the brain of P21 WT and *Gba*$^{\Delta Nes}$ mice. (D) Body weight chart for WT, *Clec4e*$^{-/-}$, *Gba*$^{\Delta Nes}$, and *Gba*$^{\Delta Nes}$ × *Clec4e*$^{-/-}$ mice (n = 15 mice per group). Error bars indicate means ± standard deviation. (E) Brain weight of P21 WT (n = 11), *Clec4e*$^{-/-}$ (n = 17), *Gba*$^{\Delta Nes}$ (n = 12), and *Gba*$^{\Delta Nes}$ × *Clec4e*$^{-/-}$ (n = 18) mice. Bars indicate the mean for each group and each dot represents an individual mouse. (F) Kaplan–Meier survival curve of WT (n = 10), *Clec4e*$^{-/-}$ (n = 10), *Gba*$^{\Delta Nes}$ (n = 10) and *Gba*$^{\Delta Nes}$ × *Clec4e*$^{-/-}$ mice (n = 21). (G) Representative FACS plots showing the gates for microglia, T and B cells, neutrophils, macrophages, and dendritic cells. (H) Representative FACS plots showing the gates for astrocytes, oligodendrocytes, and neurons. (I) Schematic view of the generation of *Clec4e*$^{\Delta Nes}$ mice. Black bars represent *Clec4e* exons, open triangles represent loxP sites, gray triangles represent FRT sites, and arrow heads denote primer location. (J) PCR products showing removal of the exons 3 and 4 of *Clec4e* in the microglia and neurons from *Clec4e*$^{\Delta Cx3cr1}$ and *Clec4e*$^{\Delta Nes}$ mice. (K) Western blot analysis of Mincle expression in the primary microglia from *Clec4e*$^{flox/flox}$ and *Clec4e*$^{\Delta Cx3cr1}$ mice stimulated with lipopolysaccharide (LPS; 1 μg/mL) overnight. (L) Schematic view of generation of "double *cre* × double flox" genetic model. Deficiency of GCase in microglia (WT vs. *Gba*$^{flox/flox}$ × *Cx3cr1-cre*; *Gba*$^{\Delta Cx3cr1}$ or *Gba*$^{\Delta Nes}$ vs. *Gba*$^{\Delta Nes}$ × *Cx3cr1-cre*; *Gba*$^{\Delta Nes\Delta Cx3cr1}$) and Mincle in neuron (*Gba*$^{\Delta Nes}$ vs. *Gba*$^{\Delta Nes}$ × *Clec4e*$^{flox/flox}$; *Gba*$^{\Delta Nes}$ × *Clec4e*$^{\Delta Nes}$) did not change disease. Deficiency of Mincle in microglia (*Gba*$^{\Delta Nes}$ vs. *Gba*$^{\Delta Nes\Delta Cx3cr1}$ × *Clec4e*$^{\Delta Nes\Delta Cx3cr1}$) ameliorated disease. Data are representative of two (B, C, J, and K) or at least three independent experiments (G and H). The two-way repeated measures ANOVA followed by Fisher's LSD test (D), the one-way ANOVA followed by Tukey–Kramer test (E) and the log-rank test (F) were used for statistical analysis. ***$p < 0.001$; ****$p < 0.0001$.

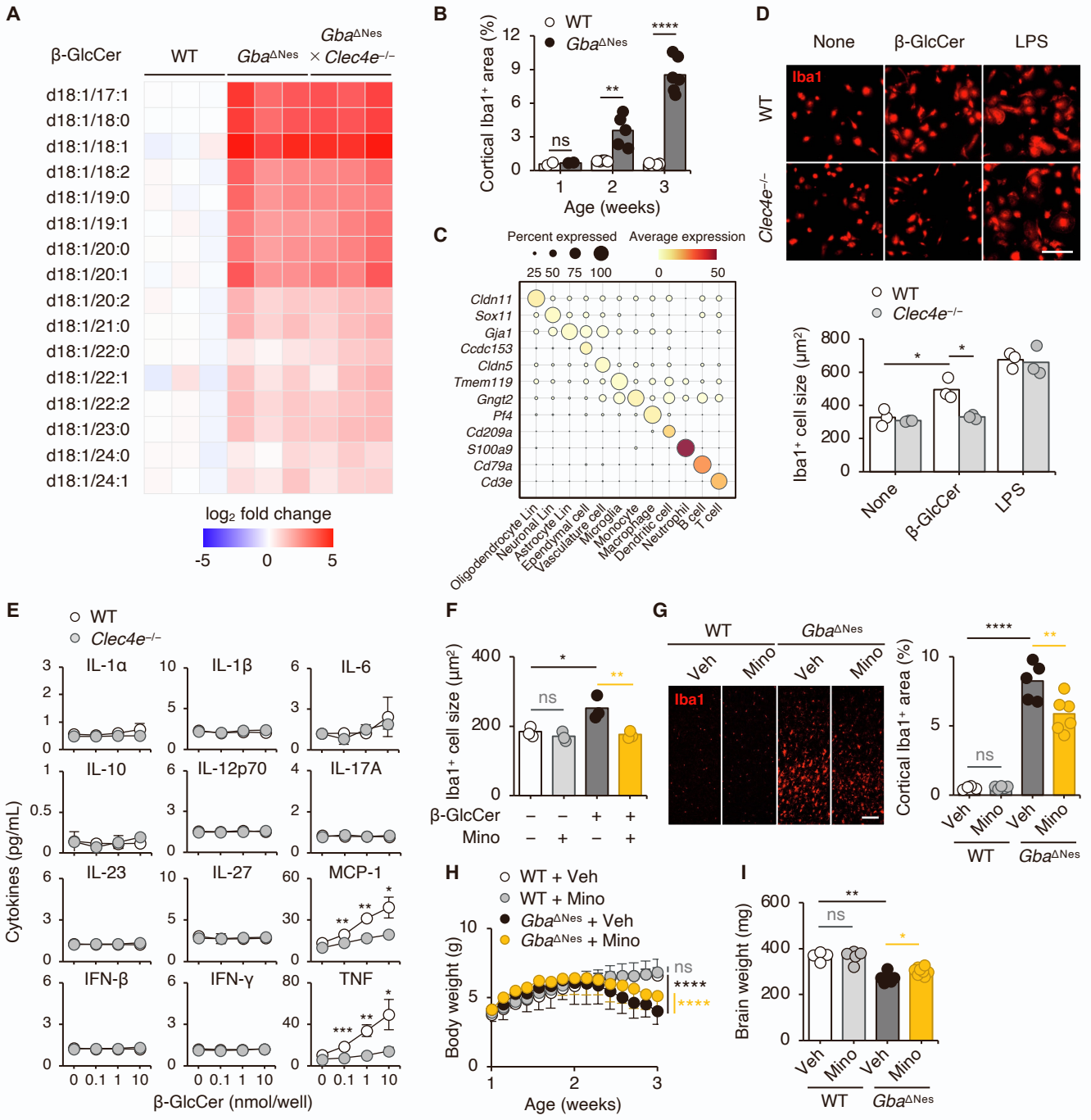


Figure S2. Activation status of microglia in mouse GD models (Related to Figure 2)

(A) Heatmap showing $\log_2$ fold change in the amount of β-GlcCer species in the brain of P14 *Gba*$^{\Delta Nes}$ and *Gba*$^{\Delta Nes}$ × *Clec4e*$^{-/-}$ mice compared to P14 WT mice (n = 3 mice per group). (B) Iba1^{+} area of the cerebral cortex of WT and *Gba*$^{\Delta Nes}$ mice at indicated age (WT, n = 2, 5, and 5; *Gba*$^{\Delta Nes}$, n = 2, 5, and 6 for 1, 2, and 3 weeks). (C) Dot plot showing the marker genes for cell types with their percent expressed and average expression. (D) Primary microglia from WT and *Clec4e*$^{-/-}$ mice were cultured with β-GlcCer d18:1/18:0 (10 nmol/well) and LPS (1 μg/mL) overnight and stained with Iba1 (red). Iba1^{+} cell size from three technical replicates. (E) Primary microglia from WT and *Clec4e*$^{-/-}$ mice were cultured overnight with the indicated amount of β-GlcCer d18:1/18:0. Concentration of cytokines in the supernatant from three technical replicates. (F) Primary microglia from WT mice were cultured overnight with or without β-GlcCer d18:1/18:0 (10 nmol/well) and/or Mino (50 μg/mL). Iba1^{+} cell size from three technical replicates. (G) Iba1 (red) staining of cerebral cortex from P21 *Gba*$^{\Delta Nes}$ mice with (WT, n = 5; *Gba*$^{\Delta Nes}$, n = 6) or without (WT, n = 4; *Gba*$^{\Delta Nes}$, n = 5) Mino treatment (left). Iba1^{+} area was quantified (right). (H) Body weight chart for WT and *Gba*$^{\Delta Nes}$ mice with or without Mino treatment (n = 5 mice per group). (I) Brain weight of P21 WT and *Gba*$^{\Delta Nes}$ mice with (WT, n = 5; *Gba*$^{\Delta Nes}$, n = 8) or without (WT, n = 4; *Gba*$^{\Delta Nes}$, n = 6) Mino treatment. Scale bars show 30 μm (D), and 100 μm (G). Each dot represents an individual mouse (B, G, and I), bars represent the mean of each group (B, D, F, G, and I), and error bars indicate mean ± standard deviation (E and H). Data are representative of three independent experiments (D and F) and at least three mice per group (G). The one-way ANOVA followed by Tukey–Kramer test (D, F, G, and I), the unpaired two-tailed Student *t*-test (B and E), and the two-way repeated measures ANOVA followed by Fisher's LSD test (H) were used for statistical analysis. *p < 0.05; **p < 0.01; ****p < 0.0001; ns, not significant.

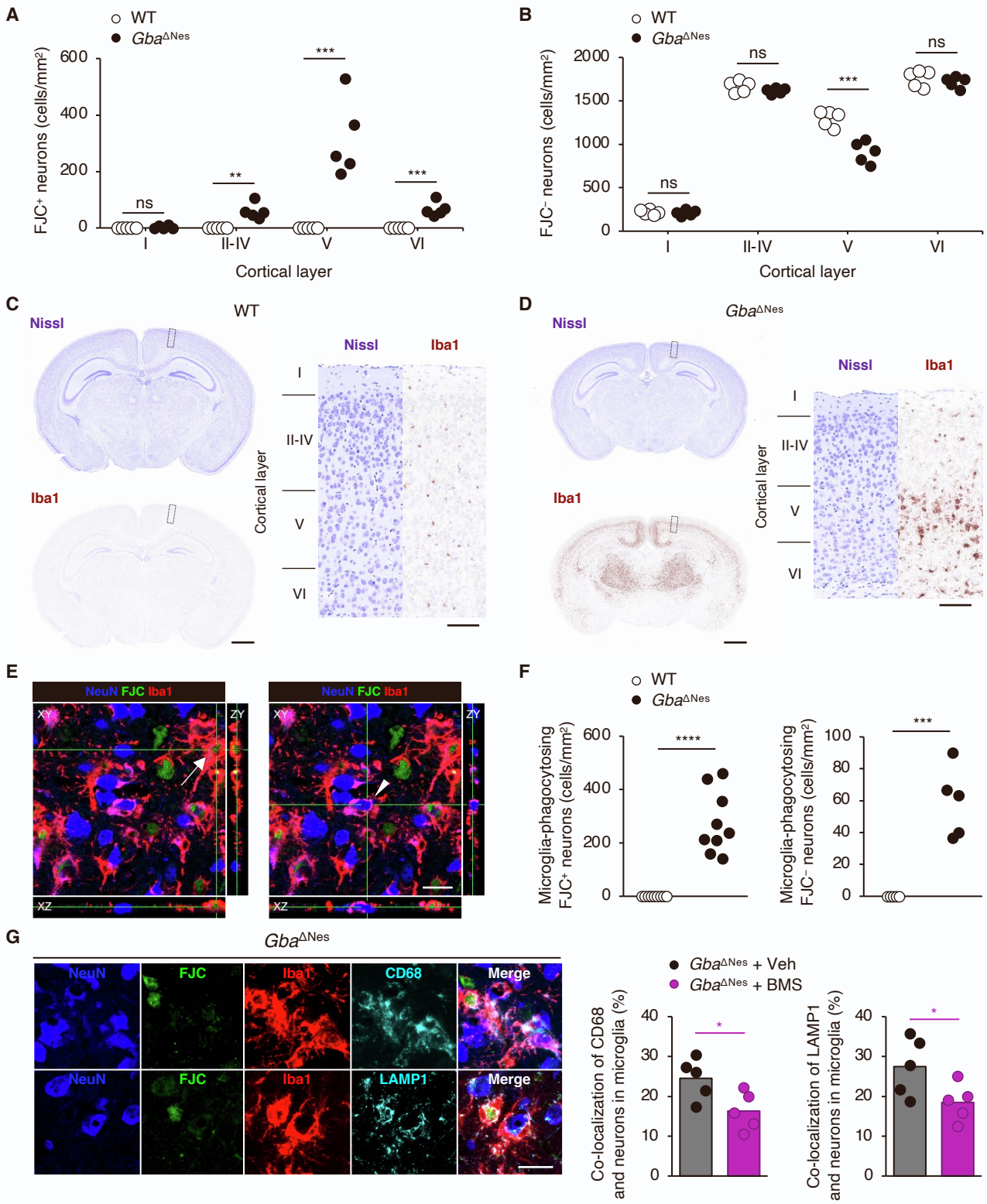


Figure S3. Histological analysis of neuron in *Gba*$^{\Delta Nes}$ mice (Related to Figure 3)

(A) Number of FJC$^+$ neurons of P21 WT and *Gba*$^{\Delta Nes}$ mice in indicated cortical layers (n = 5 mice per group). (B) Number of FJC$^-$ neurons of P21 WT and *Gba*$^{\Delta Nes}$ mice in indicated cortical layers (n = 5 mice per group). (C) Coronal section from P21 WT mice stained by Iba1 with Nissl counterstain (left). High magnification images showing cortical layers (right). (D) Coronal section from P21 *Gba*$^{\Delta Nes}$ mice stained by Iba1 with Nissl counterstain (left). High magnification images showing cortical layers (right). (E) Cortical layer V from P21 *Gba*$^{\Delta Nes}$ mice stained with NeuN (blue), FJC (green), and Iba1 (red) in XY, XZ, and ZY projections. Arrowhead and arrow indicate living neurons (FJC$^-$NeuN$^+$) and degenerating neurons (FJC$^+$) phagocytosed by microglia, respectively. Analysis was performed on a minimum of 5 mice per group, representative image shown. (F) Number of microglia-phagocytosing FJC$^+$ degenerating neurons (n = 9 mice per group) (left) and FJC$^-$ living neurons (n = 5 mice per group) (right). (G) Cortical layer V from P21 *Gba*$^{\Delta Nes}$ mice stained with NeuN (blue), FJC (green), Iba1 (red), and CD68 or LAMP1 (cyan) (left). Percentage of co-localization of CD68 or LAMP1 and neurons in microglia of Veh or BMS-treated *Gba*$^{\Delta Nes}$ mice (n = 5 mice per group) (right). Scale bars show 1 mm (C and D left panel), 100 μm (C and D right panel), and 20 μm (E and G). Each dot represents an independent mouse (A, B, F, and G) and bars are the mean of each group (G). Data were analyzed by the unpaired two-tailed Student *t*-test (A, B, F, and G). *p < 0.05; **p < 0.01; ***p < 0.001; ****p < 0.0001; ns, not significant.

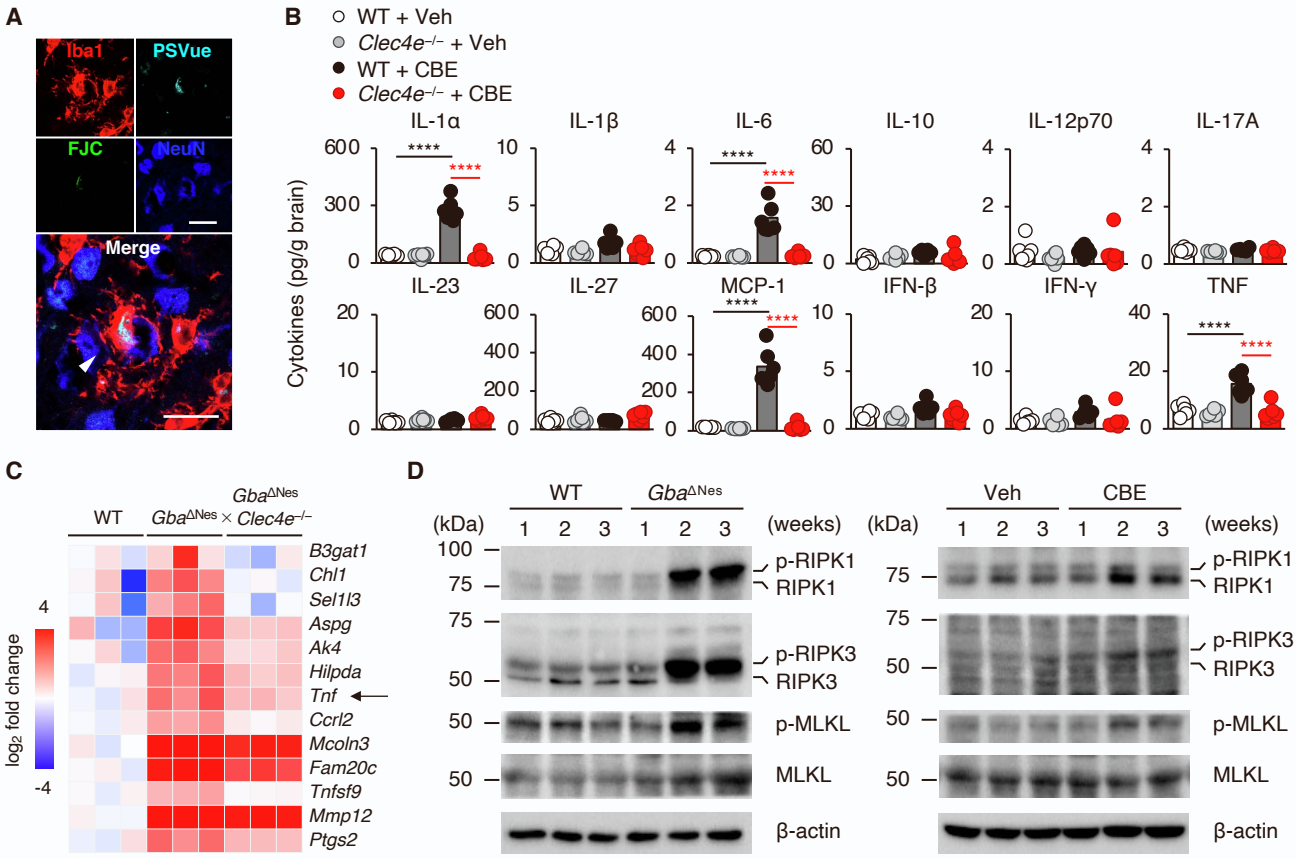


Figure S4. Inflammatory signature in the brain of GD mice (Related to Figure 4)

(A) Cortical layer V from P21 *Gba*$^{\Delta Nes}$ mice stained with Iba1 (red), PSVue (cyan), FJC (green), and NeuN (blue). Arrowhead indicates PtdSer-exposing living neuron (PSVue^{+}FJC^{-}NeuN^{+}) phagocytosed by microglia. Scale bars show 20 μm. (B) Inflammatory cytokines in the brain from P21 WT and *Clec4e*$^{-/-}$ mice with or without CBE treatment (n = 6 mice per group). Each dot represents an independent mouse and bars are the mean for each group. (C) Bulk RNA-seq of isolated microglia from P14 WT and *Gba*$^{\Delta Nes}$ mice brain. Heatmap showing $\log_2$ fold change of all downregulated DEGs ($p < 0.05$, $\log_2$ fold change ≥ 2) between *Gba*$^{\Delta Nes}$ vs. *Gba*$^{\Delta Nes}$ × *Clec4e*$^{-/-}$ microglia among upregulated DEGs ($p < 0.05$, $\log_2$ fold change ≥ 2) between WT vs. *Gba*$^{\Delta Nes}$ microglia (n = 3 mice per group). Arrow indicates *Tnf*. (D) Western blot analysis of phosphorylated and non-phosphorylated RIPK1, RIPK3, and MLKL expression in neurons from a WT and *Gba*$^{\Delta Nes}$ or Vehicle and CBE-treated WT mice at the indicated age. Data are representative of two mice (A) and two independent experiments (D). The one-way ANOVA followed by Tukey–Kramer test (B) was used for statistical analysis. ****$p < 0.0001$.

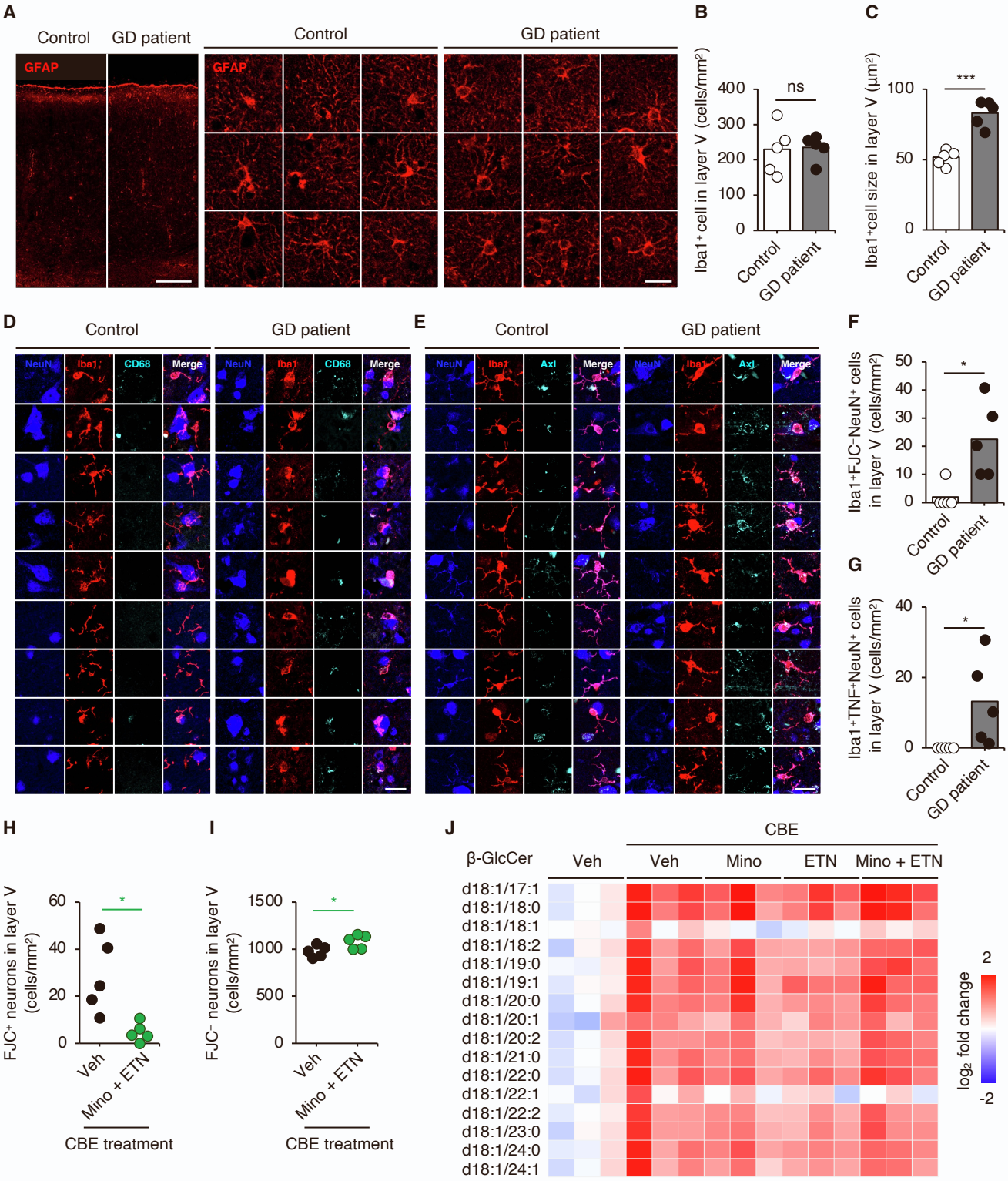


Figure S5. Effect of combinatory use of FDA-approved drugs on neuronal loss (Related to Figure 5)

(A) Images of GFAP+ cells in cortical layer V of control and GD patients. (B) Number of Iba1+ cells in cortical laver V of control and GD patients. (C) Size of Iba1+ cells in cortical layer V of control and GD patients. (D) Images of NeuN (blue), Iba1 (red), and CD68 (cyan) cells in the cortex of control and GD patients. (E) Images of NeuN (blue), Iba1 (red), and Axl (cyan) cells in the cortex of control and GD patients. (F) Number of Iba1+ FJC- NeuN+ cells in cortical layer V of control and GD patients. (G) Number of Iba1+ TNF+ NeuN+ cells in cortical layer V of control and GD patients. (H) Number of FJC+ neurons in layer V of P21 WT mice treated with CBE or CBE + Mino + ETN (n = 5 mice per group). (I) Number of FJC- neurons in layer V of P21 WT mice treated with CBE or CBE + Mino + ETN (n = 5 mice per group). (J) Heatmap showing $\log_2$ fold change in the amount of β-GlcCer species in the brain of P21 CBE mice treated with Veh, Mino, ETN, or Mino + ETN compared with P21 Vehicle-treated WT mice (n = 3 mice per group). Each dot represents an individual patient image (B, C, F, and G) or mouse (H and I). Bars are the mean for each group (B, C, F and G). Scale bars are 500 μm (A left panels) and 20 μm (A right panels, D, and E). Data were analyzed by the unpaired two-tailed Student *t*-test (B, C, and F–I). *$p < 0.05$; ***$p < 0.001$; ns, not significant.